

Statistics and Probability:
Naïve Bayes

Applications of Naive Bayes Algorithm

- As you must've noticed, this algorithm offers plenty of advantages to its users. That's why it has a lot of applications in various sectors too. Here are some applications of the Naïve Bayes algorithm:
  - As this algorithm is fast and efficient, you can use it to make real-time predictions.
  - This algorithm is popular for multi-class predictions. You can find the probability of multiple target classes easily by using this algorithm.
  - Email services (like Gmail) use this algorithm to determine whether an email is a spam. This algorithm is excellent for spam filtering.
  - Its assumption of feature independence, and its effectiveness in solving multi-class problems, makes it perfect for performing Sentiment Analysis. Sentiment Analysis refers to identifying a target group's positive or negative sentiments (customers, audience, etc.)
  - Collaborative Filtering and the Naive Bayes algorithm work together to build recommendation systems. These systems use data mining and machine learning to predict whether the user would like a particular resource.

Note: Naive Bayes is a specific type of Bayesian classifier, but not all Bayesian classifiers are Naive Bayes classifiers.

Basic Probability:

Experimental Probability

Experimental Probability is found by repeating an experiment and observing the outcomes.

$$P(\text{event}) = \frac{\text{number of times event occurs}}{\text{total number of trials}}$$

Example:

A coin is tossed 10 times.
A head is recorded 7 times and a tail 3 times.

$$P(\text{head}) = \frac{7}{10} \qquad P(\text{tail}) = \frac{3}{10}$$

Simple Probability

$$\text{Probabilty} = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$$



Example:

$$P(red) = \frac{7}{12}$$

← Number of red marbles

← Total number of marbles (sample space)

$$P(blue) = \frac{5}{12}$$

← Number of blue marbles

← Total number of marbles (sample space)

Bayesian Classifier: Notation

$\boldsymbol{x} \in \mathbb{R}^d$: d-dimensional feature vector

y : class number
(usually $y \in \{0, 1\}$ or $y \in \{-1, +1\}$)

$p(y)$: prior probability of pattern class y

$p(\boldsymbol{x})$: evidence
(distribution of features in d-dimensional feature space)

$p(\boldsymbol{x}, y)$: joint probability density function (pdf)

$p(\boldsymbol{x}|y)$: class conditional density

$p(y|\boldsymbol{x})$: posterior probability

Bayesian classifiers have been used in a wide range of applications, including email spam filtering, medical diagnosis, image recognition, and natural language processing.

Bayesian Classifier: Application

For the following observed counts of rolling either dice 1 (D1) or dice 2 (D1) multiple times:



	1	2	3	4	5	6
D1	9	12	7	11	7	11
D2	7	8	9	7	8	12

Bayesian Classifier: Application

where $y \in \{D1, D2\}$ and $x \in \{1, 2, 3, 4, 5, 6\}$.

(a) Estimate $P(x = 4)$. (11+7)/108

(b) Estimate $P(y = D2)$. 51/108

(c) Estimate $P(x = 4, y = D1)$. 11/108

(d) Estimate $P(x = 4|y = D2)$. 7/51

Bayesian Classifier

Bayesian Classifier

$$\underbrace{p(\boldsymbol{x}, y)}_{\text{joint pdf}} = \underbrace{p(y)}_{\text{prior}} \cdot \underbrace{p(\boldsymbol{x}|y)}_{\text{class conditional pdf}}$$

Bayesian Classifier

$$\underbrace{p(\boldsymbol{x}, y)}_{\text{joint pdf}} = \underbrace{p(y)}_{\text{prior}} \cdot \underbrace{p(\boldsymbol{x}|y)}_{\text{class conditional pdf}}$$

$$= \underbrace{p(\boldsymbol{x})}_{\text{evidence}} \cdot \underbrace{p(y|\boldsymbol{x})}_{\text{posterior}}$$

Bayesian Classifier

Now we get the posterior as follows:

$$p(y|\boldsymbol{x}) = \frac{p(y) \cdot p(\boldsymbol{x}|y)}{p(\boldsymbol{x})}$$

Bayesian Classifier: Optimality

Now let us summarize the Bayesian decision rule:

We decide for the class y^* according to the decision rule

$$
\begin{aligned}
y^* &= \underset{y}{\operatorname{argmax}}\, p(y|\boldsymbol{x}) \\
&= \underset{y}{\operatorname{argmax}}\, \frac{p(y) \cdot p(\boldsymbol{x}|y)}{p(\boldsymbol{x})} \\
&= \underset{y}{\operatorname{argmax}}\, p(y) \cdot p(\boldsymbol{x}|y) \\
&= \underset{y}{\operatorname{argmax}} \{\log p(y) + \log p(\boldsymbol{x}|y)\}
\end{aligned}
$$

Bayesian Classifier: Optimality

Using the $(0,1)$-loss function, the class decision is based on:

$$
\begin{aligned}
y^* &= \operatorname*{argmin}_y \mathrm{AL}(\boldsymbol{x}, y) \\
&= \operatorname*{argmin}_y \sum_{y'} l(y, y') \cdot p(y'|\boldsymbol{x}) \\
&= \operatorname*{argmax}_y p(y|\boldsymbol{x})
\end{aligned}
$$

Naïve Bayes: Problems

Bayesian Classifier: Problem

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Bayesian Classifier: Problem

Dataset

Color	Type	Origin	Stolen?
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes

Frequency Table

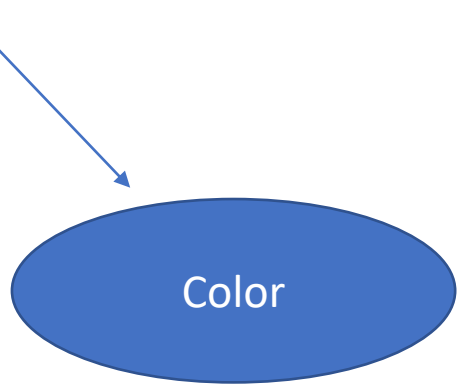
		Stolen?	
		Yes	No
Color	Red	3	2
	Yellow	2	3

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Color	Red	3/5	2/5
	Yellow	2/5	3/5

Bayesian Classifier: Problem

Color	Type	Origin	Stolen?
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
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Frequency Table

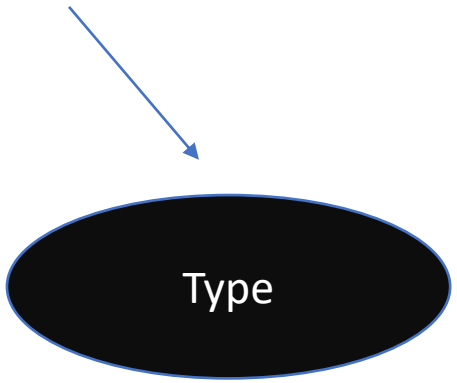
		Stolen?	
		Yes	No
Color	Red	3	2
	Yellow	2	3

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Color	Red	3/5	2/5
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Bayesian Classifier: Problem

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Red	Sports	Domestic	Yes
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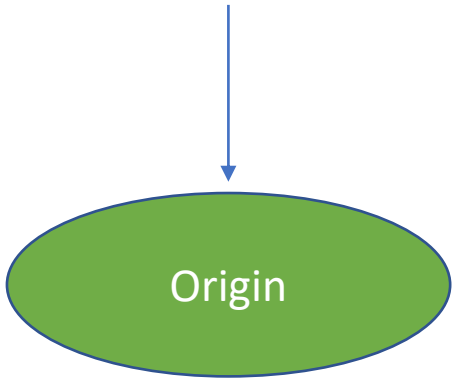
		Stolen?	
		Yes	No
Type	Sports	4	2
	SUV	1	3

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Type	Sports	4/5	2/5
	SUV	1/5	3/5

Bayesian Classifier: Problem

Color	Type	Origin	Stolen?
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes



Frequency Table

		Stolen?	
		Yes	No
Origin	Domestic	2	3
	Imported	3	2

Likelihood Table

		Stolen?	
		P(Yes)	P(No)
Origin	Domestic	2/5	3/5
	Imported	3/5	2/5

What is the prediction?

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

What is the prediction?

Now, The Question is -

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

As per the equations discussed above, we can calculate the posterior probability P(Yes | X) as :

$$P(Yes \mid X) = P(Red \mid Yes) * P(SUV \mid Yes) * P(Domestic \mid Yes) * P(Yes)$$

$$= 3/5 * 1/5 * 2/5 * 0.5$$
$$= 0.0092$$

What is the prediction?

Now, The Question is -

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

As per the equations discussed above, we can calculate the posterior probability P(Yes | X) as :

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$$= 3/5 * 1/5 * 2/5 * 0.5$$

$$= 0.0092$$

and P(No | X):

$$P(No \mid X) = P(Red \mid No) * P(SUV \mid No) * P(Domestic \mid No) * P(No)$$

$$= 2/5 * 3/5 * 3/5 * 0.5$$

$$= 0.072$$

What is the prediction?

Now, The Question is -

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

P(No) >> P(Yes)
Output: No

As per the equations discussed above, we can calculate the posterior probability P(Yes | X) as :

$$P(Yes \mid X) = P(Red \mid Yes) * P(SUV \mid Yes) * P(Domestic \mid Yes) * P(Yes)$$

$$= 3/5 * 1/5 * 2/5 * 0.5$$

$$= 0.0092$$

and P(No | X):

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$$= 2/5 * 3/5 * 3/5 * 0.5$$

$$= 0.072$$